

Rough slab under beam illumination: scoping the extension of the pysie2d boundary integral method

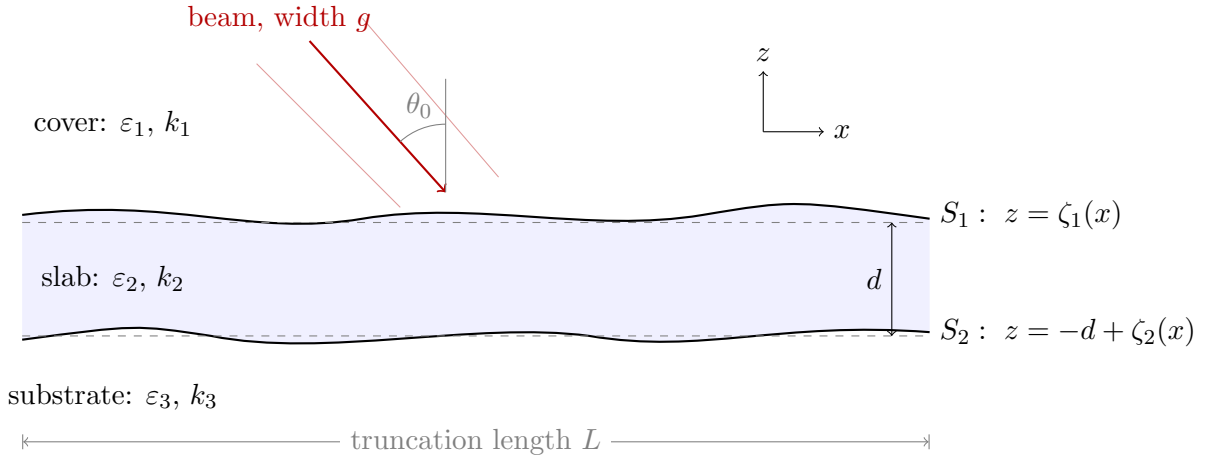
pysie2d design notes

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Abstract

This note works out what it takes to apply Maradudin’s rough-surface method (Maradudin *et al.*, Ann. Phys. 203, 255, 1990) to a film: two rough, flat-on-average surfaces extending to infinity, with a laterally bounded Gaussian beam coming from above. Starting from the boundary conditions, it derives the four coupled integral equations and shows that they have the same building blocks as the pysie2d matrix: each surface contributes a copy of the single-particle 2×2 block, and the two surfaces are coupled through the slab. The one new difficulty is that truncating infinite surfaces fails once roughness couples light into guided modes, which carry power sideways to the cut ends. The flat slab has an exact closed-form answer, and it cannot excite guided modes, so it is a clean first validation anchor. Rough-surface physics also needs a small-roughness perturbation anchor.

1 Geometry, media and conventions



Everything is invariant along y , so, as in pysie2d, a single scalar field $\psi(x, z)$ describes each polarisation:

- pol = 2, TE: $\psi = E_y$;
- pol = 1, TM: $\psi = H_y$.

The time dependence is $e^{-i\omega t}$ and lengths are in nm. Each medium j has a complex permittivity ε_j and wavenumber $k_j = \sqrt{\varepsilon_j} 2\pi/\lambda_{\text{vac}}$. In every medium the field satisfies $\nabla^2 \psi + k_j^2 \psi = 0$, and the free-space Green function is the one pysie2d already uses:

$$G_j(r) = \frac{i}{4} H_0^{(1)}(k_j r). \quad (1)$$

Only the slab is allowed to be lossy. The cover and substrate must be lossless so that far-field reflection and transmission exist (see §6).

Surface unknowns. As in Maradudin's paper, both surfaces are written as height profiles, parametrised by x and sampled over $-L/2 < x < L/2$. On each surface there are two unknown functions:

- $\phi_m(x)$: the field on surface S_m ;
- $\chi_m(x)$: the derivative along the *un-normalised* upward normal, $\chi = (-\zeta'_m \partial_x + \partial_z) \psi$.

Because χ carries the length element $ds = \sqrt{1 + \zeta'^2} dx$, every surface integral below reduces to a plain $\int dx'$. This is the same idea as pysie2d's χ carrying the Jacobian of t (conventions §5). On S_1 , χ_1 is taken on the cover side; on S_2 , χ_2 is taken on the slab side.

2 Boundary conditions

On each surface, the tangential E and H are continuous. For the scalar field this becomes:

$$\psi \text{ continuous}, \quad \frac{1}{\varepsilon^p} \frac{\partial \psi}{\partial n} \text{ continuous}, \quad p = \begin{cases} 0 & \text{TE} \\ 1 & \text{TM} \end{cases} \quad (2)$$

So the field is the same on both sides of a surface, and the normal derivative jumps by a known ratio:

$$\text{slab side of } S_1 = a_{12} \chi_1, \quad \text{substrate side of } S_2 = a_{23} \chi_2, \quad a_{12} = \left(\frac{\varepsilon_2}{\varepsilon_1}\right)^p, \quad a_{23} = \left(\frac{\varepsilon_3}{\varepsilon_2}\right)^p. \quad (3)$$

For TE, $a = 1$. For TM, a is the ratio of permittivities, which is the same factor pysie2d applies as `eta = eps` to its interior block.

3 From Green's theorem to four coupled equations

Apply Green's second identity in each of the three regions, taking the observation point onto a surface (the usual half-value jump of the double-layer term). This gives two equations per surface: one from the medium above it and one from the medium below. We need two kernels for points $(x, \zeta(x))$ and $(x', \zeta(x'))$, with separation $R = \sqrt{(x - x')^2 + (\zeta(x) - \zeta(x'))^2}$:

$$\text{single-layer: } G_j(x, x') = \frac{i}{4} H_0^{(1)}(k_j R), \quad (4)$$

$$\text{double-layer: } D_j(x, x') = \frac{ik_j}{4} H_1^{(1)}(k_j R) \frac{[\zeta(x) - \zeta(x')] - (x - x') \zeta'(x')}{R}. \quad (5)$$

When $x' \rightarrow x$, G_j has a logarithmic singularity, while D_j tends to the finite value $\zeta''/(4\pi(1 + \zeta'^2))$. These are the same two singular behaviours that pysie2d's Kress quadrature already handles.

With $\int_{S_m}$ meaning $\int_{-L/2}^{L/2} dx'$ along surface m , the four equations are:

$$\text{cover, on } S_1 : \quad \frac{1}{2} \phi_1 - \int_{S_1} [D_1 \phi_1 - G_1 \chi_1] = \psi_{\text{inc}} \quad (6)$$

$$\text{slab, on } S_1 : \quad \frac{1}{2} \phi_1 + \int_{S_1} [D_2 \phi_1 - a_{12} G_2 \chi_1] - \int_{S_2} [D_2 \phi_2 - G_2 \chi_2] = 0 \quad (7)$$

$$\text{slab, on } S_2 : \quad \frac{1}{2} \phi_2 + \int_{S_1} [D_2 \phi_1 - a_{12} G_2 \chi_1] - \int_{S_2} [D_2 \phi_2 - G_2 \chi_2] = 0 \quad (8)$$

$$\text{substrate, on } S_2 : \quad \frac{1}{2} \phi_2 + \int_{S_2} [D_3 \phi_2 - a_{23} G_3 \chi_2] = 0 \quad (9)$$

Equations (7) and (8) are the same slab equation, evaluated on the upper and lower surface respectively. The slab is the only region bounded by both surfaces, and it is the only place the surfaces talk to each other.

Matrix structure. Sample each surface at N points, so there are $4N$ unknowns $[\phi_1, \chi_1, \phi_2, \chi_2]$:

$$\left[\begin{array}{cc|cc} \frac{1}{2} - D_1 & G_1 & 0 & 0 \\ \frac{1}{2} + D_2 & -a_{12}G_2 & -D_2 & G_2 \\ \hline D_2 & -a_{12}G_2 & \frac{1}{2} - D_2 & G_2 \\ 0 & 0 & \frac{1}{2} + D_3 & -a_{23}G_3 \end{array} \right] \begin{bmatrix} \phi_1 \\ \chi_1 \\ \phi_2 \\ \chi_2 \end{bmatrix} = \begin{bmatrix} \psi_{\text{inc}} \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

- The **diagonal 2×2 blocks** have the same structure as the `pysie2d` single-particle matrix: one row for the medium above, one for the medium below, with a multiplying the lower medium's single layer. On S_1 the pair is (cover, slab). On S_2 it is (slab, substrate).
- The **off-diagonal blocks** couple S_1 and S_2 through the slab only, and they are smooth because the surfaces never touch. They play the role of `assemble_cross_block` in `cluster.py`, except that the coupling medium is the slab, not the background.
- **Orientation.** `pysie2d`'s normal points to the right of the direction of travel (outward on a counter-clockwise curve). Traversing each surface from right to left makes that normal point up, so the upper medium plays the role of the particle's "exterior". This has to be written into `conventions.md`.

The dropped end segments. Green's theorem in the slab really runs around a closed contour. The truncated equations keep the two long sides and drop the two short vertical ends at $x = \pm L/2$. That is harmless when the fields there are negligible. It fails exactly when light is guided along the slab (§5).

4 The incident beam

Following Maradudin, the beam is a sum of plane waves with a Gaussian weight, restricted to propagating directions so that it carries finite power:

$$\psi_{\text{inc}}(x, z) = \int_{-k_1}^{k_1} dq W(q) e^{iqx - i\alpha_1(q)z}, \quad \alpha_1 = \sqrt{k_1^2 - q^2}, \quad W(q) = \frac{g}{2\sqrt{\pi}} e^{-g^2(q - k_1 \sin \theta_0)^2/4}. \quad (10)$$

At normal incidence its profile on $z = 0$ is close to e^{-x^2/g^2} . The q -integral is smooth and is done by ordinary quadrature at each surface point. It fills only the ϕ_1 rows, just as `pysie2d`'s plane wave fills only the ϕ half.

Sizing rules:

- $k_1 g \cos \theta_0 \gg 1$ (in practice $g \gtrsim 3\lambda$), otherwise cutting off the evanescent part of the spectrum distorts the beam;
- the footprint is $g/\cos \theta_0$. For the beam amplitude at $x = \pm L/2$ to be $\leq 10^{-6}$, $L \approx 7.5 g/\cos \theta_0$. Near grazing incidence L grows without bound;
- near a leaky resonance the reflected beam is shifted sideways by the leakage length (Goos-Hänchen), and L must include that shift.

5 Truncation and guided modes: the one new problem

A single rough surface (Maradudin's case) is safe to truncate: every part of the field decays away from the beam footprint. A slab is different:

- **Flat slab: no problem.** A propagating beam from the cover has $|q| < k_1$, while a guided mode needs $q > \max(k_1, k_3)$. The flat slab therefore cannot excite guided modes, the ends stay dark, and truncation is clean.

- **Rough slab: problem.** Roughness supplies the missing lateral momentum, and it couples part of the beam into guided modes, which travel along x without decaying if the slab is lossless. They reach $x = \pm L/2$, where they meet the missing end segments. That acts like a cleaved facet: the mode reflects back and creates spurious interference. The ensemble average does not remove it.

The options, from cheapest to most complete:

1. **Loss in the slab.** Choose $\text{Im } \varepsilon_2$ so that the mode decays by the tolerance over the distance from footprint to end. This is trivial to implement, and complex ε already works everywhere. But it changes the physics: lossless answers need an extrapolation $\text{Im } \varepsilon_2 \rightarrow 0$, and guided-mode interference effects are damped.
2. **Complex-stretched ends (a perfectly matched layer inside the integral equation).** Beyond the footprint, give the surface coordinate an imaginary part, $x \rightarrow x + i\sigma(x)$, so outgoing guided waves decay without reflecting. The Hankel functions then get complex arguments. `pysie2d` supports those by design (non-negotiable 1), so the kernels need no change. Prior work: Lu, Lu and Qian, *SIAM J. Appl. Math.* 78 (2018), PML boundary integral method for layered media.
3. **Windowed Green function** (Bruno, Lyon, Pérez-Arancibia, Turc, *Proc. R. Soc. A* 2017, layered-media version). A smooth window multiplies the integrands. Its behaviour with guided modes must be checked against the paper before relying on it.
4. **Closing the ends explicitly** with the known guided-mode fields. This is the most work, and it is only worth it if the guided power itself is the observable.

Recommendation: option 1 for the first rough-slab results, then option 2 once lossless slabs matter. Energy conservation only closes if the power that leaves through the ends (or is absorbed in the stretched region) is counted as a separate term.

6 Quadrature on open surfaces

`pysie2d`'s Kress rule assumes a closed, periodic curve. A truncated surface is not closed. There are two ways forward:

- **Periodic Kress on the truncated interval.** Map $x \in [-L/2, L/2)$ to $t \in [0, 2\pi)$ and use the existing weights unchanged. The two ends are then treated as neighbours. The spurious end-to-end term is multiplied by the surface fields at the ends, which are at the beam-tail level (flat slab), or suppressed by loss or stretching (rough slab). The expected error floor is therefore the tail amplitude. That is a claim to measure, not to assume (Gate 0).
- **Fallback:** Maradudin's original rule, which is midpoint sampling with the log singularity integrated analytically on the diagonal. It is first order, like pre-v0.6 `pysie2d`. This is acceptable for ensemble statistics where Monte Carlo noise dominates, but a step down from the package's standard.

Sampling constraints:

- $\Delta x \lesssim \lambda/(10 n_{\text{max}})$ for the first-order rule; fewer points with Kress;
- $\Delta x \ll a$, the roughness correlation length, with $a/\Delta x \gtrsim 5$;
- $\Delta x \ll d$: for a thin slab the coupling blocks become nearly singular, the same issue as nearly touching particles in `cluster.py`.

7 Observables

Far-field amplitudes. Far above the slab, the scattered field is an outgoing cylindrical wave times the amplitude

$$r(\theta_s) = \int_{S_1} dx' e^{-ik_1(x' \sin \theta_s + \zeta_1(x') \cos \theta_s)} \left[-ik_1(\cos \theta_s - \zeta'_1 \sin \theta_s) \phi_1 - \chi_1 \right]. \quad (11)$$

The transmitted amplitude $t(\theta_t)$ has the same form: it uses surface S_2 , k_3 , downward directions and the substrate-side derivative $a_{23}\chi_2$. This is `pysie2d`'s existing far-field formula applied to an open curve. It only exists if the cover and substrate are lossless.

Normalisation. The power scattered per unit angle is $|r|^2/8\pi$, and the beam brings in $2\pi \int |W(q)|^2 \alpha_1(q) dq$. The polarisation-dependent flux factor cancels in the ratio, so

$$\frac{\partial R}{\partial \theta_s} = \frac{|r(\theta_s)|^2}{16\pi^2 \int |W(q)|^2 \alpha_1(q) dq}, \quad (12)$$

The same holds for transmission, with an extra $\varepsilon_1/\varepsilon_3$ in TM because the TM flux carries $1/\varepsilon$.

Energy balance. $R + T + A + P_{\text{ends}} = 1$, where A is absorption in the slab (this connects to `absorption-scope.md`) and P_{ends} is the guided power lost at the ends.

Ensemble statistics. Each rough realisation, generated by filtering white noise with the roughness spectrum (Gaussian correlation, rms height δ , correlation length a , using numpy FFT only), gives one $r(\theta_s)$. Averaging over realisations gives

- the coherent part, $|\langle r \rangle|^2$;
- the incoherent part, $\langle |r|^2 \rangle - |\langle r \rangle|^2$.

The latter shows the enhanced backscattering peak in the retroreflection direction and, for a film, satellite peaks.

8 Validation anchors

Per the package rule, new physics needs an anchor that is not a second path in this repository.

G0 Open-surface quadrature. On the flat slab, measure how the boundary-value error falls with N and with L against the closed form below. Pass: it reaches the beam-tail floor. This decides §6.

G1 Flat slab, closed form. Each plane-wave component reflects with the Airy/Fresnel film coefficient

$$r_{\text{slab}}(q) = \frac{r_{12} + r_{23}e^{2i\alpha_2 d}}{1 + r_{12}r_{23}e^{2i\alpha_2 d}}, \quad r_{ij} = \frac{\alpha_i/\varepsilon_i^p - \alpha_j/\varepsilon_j^p}{\alpha_i/\varepsilon_i^p + \alpha_j/\varepsilon_j^p}, \quad \alpha_j = \sqrt{k_j^2 - q^2}.$$

This gives the exact surface field, $\phi_1(x) = \int W(q) [1 + r_{\text{slab}}(q)] e^{iqx} dq$, which can be compared point by point with the solution vector, and it gives exact R , T and A . It covers both polarisations, different cover and substrate, a lossy slab, total internal reflection, and the Goos-Hänchen shift. This is the main anchor. It needs a new `reference/slab.py`, the counterpart of `reference/mie.py`.

G2 Weak roughness. First-order small-amplitude perturbation theory gives the incoherent reflection in closed form, as the flat-slab response times the roughness power spectrum. The check is agreement as $\delta \rightarrow 0$ and the δ^2 scaling.

G3 Guided-mode signatures. The flat-slab mode indices come from the poles of r_{slab} , a closed form. Rough-slab satellite peaks must appear at $\sin \theta_s = -\sin \theta_0 \pm (q_1 - q_2)/k_1$ for mode pairs (q_1, q_2) (McGurn–Maradudin; Freilikher *et al.*).

G4 Consistency checks (not independent anchors):

- setting $\varepsilon_3 = \varepsilon_2$ removes S_2 and must reproduce the single rough interface;
- reciprocity of the ensemble-averaged reflection;
- energy balance.

9 Cost

There are $4N$ unknowns, and the dense matrix takes $16(4N)^2$ bytes. The kernel work is about $3N^2$ Hankel pairs:

- four self blocks, each using the symmetry of R ;
- one full slab cross block.

LU costs about $\frac{8}{3}(4N)^3$ flops.

N per surface	unknowns	matrix	LU	example
1000	4 000	0.26 GB	~ 1 s	$L \approx 50\lambda$, $n_2 = 2$, $\lambda/20$ sampling
2500	10 000	1.6 GB	~ 20 s	$L \approx 125\lambda$
5000	20 000	6.4 GB	~ 3 min	at the limit of a laptop

This **changes the performance profile** of the package. `pysie2d` today is limited by special-function evaluation. At these sizes, the dense solve dominates. The existing reuse principle still applies, because several incidence angles on one realisation mean one factorisation and many right-hand sides. Different realisations share nothing, so the parallelism is across realisations, sized to the four performance cores. Fast multipole or iterative solvers are out of scope here.

10 What has to be built

Item	Notes	Reuse
Open-surface geometry	height profile $\zeta(x)$ on $[-L/2, L/2)$, traversed right to left	new constructor
Three-media material	cover as background, $k_{\text{bg}} = 2\pi n_1/\lambda_{\text{vac}}$; slab and substrate relative	extends §2 conventions
Surface block	pysie2d 2×2 block with the <i>upper</i> and <i>lower</i> wavenumbers as inputs, not background and core	<code>assemble_matrix</code>
Slab coupling block	smooth, slab medium	<code>assemble_cross_block</code>
Gaussian beam source	q -integral, fills ϕ_1 rows	new
End treatment	loss first, then complex stretching	kernels unchanged
$r(\theta_s), t(\theta_t), \partial R/\partial \theta_s$	open-curve far field, beam-power normalisation	<code>far_field</code>
Rough surface generator, ensemble loop	FFT filtering; threaded over realisations	threading pattern
<code>reference/slab.py</code>	Airy film + beam spectrum; perturbation theory	new anchor
<code>conventions.md</code>	orientation, χ_2 side, a_{12} , a_{23} , background medium, normalisation	—

Suggested order.

1. Flat slab end to end, G0 and G1: geometry, blocks, beam, reference.
2. Rough slab with a lossy slab: generator, ensemble, G2, G3.
3. Complex-stretched ends for lossless slabs, with energy balance including P_{ends} .

Everything is additive and sits beside `BIESolver`.

11 Open decisions

- **Which “slab” is on the roadmap.** “Slab-waveguide backgrounds” in the roadmap may mean *particles inside* a flat slab. That calls for the layered-medium Green function (Sommerfeld integrals), not truncated surfaces. The two share the flat-slab reference, but little else.
- **Quadrature standard.** Accept first-order quadrature for rough ensembles if G0 fails, or insist on spectral accuracy?
- **Correlated surfaces.** Independent roughness on the two surfaces, identical roughness (a conformal film), or both?
- **Release placement.** After v1.0, as a new milestone, or as a separate study repository.